

Problem Set 3: Working with Chicken Nuggets

For this assignment, you will use Python and Visual Studio Code to create a computer program. Submit a Python program script file with the solutions (file extension is py) in Blackboard by Sunday at 11:59 PM. Using the information provided in the course up to this point, you will use some theories established by Diophantus of Alexandria in the third century. While you may not have heard of this scholar, the world is full of evidence proving his theories. He established a polynomial equation that only utilizes integers. One such example is the Pythagorean theorem.

Script Requirements

When writing and saving your programming files ensure that you heed the instructions. For all coursework that consists of a programming file there are specific things that must be placed in the programming code. The beginning of the script file must include the assignment name, the purpose of the programming code, the date it was written or revised, and who wrote the code. These comments can be written as line comments or a commenting block. Figure 1 depicts an example of what this might look like in your file. When saving the program for the first problem, name the file **assignment_3_1.py**. The second and third problems shall be saved with the file names **assignment_3_2.py** and **assignment_3_3.py**, respectively. When you have completed and tested your script files, submit all of the script files for this assignment in Blackboard.

As it has already been established, all analysis begins with a problem to solve, a question to answer, or something to prove (you might call this a hypothesis). The following information establishes the requirements of the program and some hints for completing this work. There are three problems that you need to solve for this assignment.

Figure 1

Content Needed in Every Script File

```
""  
Assignment 3 Problem 1: quantities of chicken nuggets that fit  
within available order quantities  
September 22, 2022  
John Smith  
""
```

Note. This contains the name of the assignment (Assignment 3 Problem 1), purpose of this assignment (quantities of chicken nuggets that fit within available order quantities), date (September 22, 2022), and the name of the developer (John Smith).

Problem One

The first problem you need to solve is the whether or not the quantity requested by the user is a viable option when ordering chicken nuggets at McDonalds. Assume that you can purchase chicken nuggets in quantities of 6, 9, and 22. That means that there is no combination of options that will allow you to purchase 23 chicken nuggets. In terms of the Diophantine equation, you can think of it as $6a + 9b + 22c = n$. To achieve this objective, there are four specific requirements. Start by asking the user how many chicken nuggets they would like to order. For each possible order quantity by box size identify which combinations equivalent to n (in other words, look for what values of a , b , and c in the Diophantine equation work). If the order is possible, print a statement to the console that states how many options are available and what these options are (a console output example is shown in Figure 2). If there are no combinations that work (such as an order of 23), let them know that they cannot order the requested quantity.

Figure 2

Example of the Console Output of Problem 1

```
How many chicken nuggets would you like to order? 52
For an order size of 52, choose from the following 2 option(s):
{'Six_piece': 2, 'Nine_piece': 2, 'Twenty_two_piece': 1}
{'Six_piece': 5, 'Nine_piece': 0, 'Twenty_two_piece': 1}
```

Note. This is an example of what the expected output may look like when you run the solution to the first problem in this assignment. The text in yellow reflects the output from the program. The text in white reflects the user's input.

Problem Two

The second problem is built on the foundation of the first problem. For this problem, it has been determined that it is simply not enough to tell the user their requested size is not feasible. You need to determine the closest feasible value of n given the quantity ordered by the user. Let the user know that their requested quantity is not feasible, but let them know they have options, by providing the quantity by box size options for this alternative feasible quantity.

Problem Three

For the third problem, you're going to build on your previous work in this assignment. The cost per nugget by quantity varies. For this problem, you need to give the user the lowest cost option. Assume that the 6-piece option costs \$3, the 9-piece costs \$4, and the 22-piece costs \$9. Instead of presenting all possible options, provide the least expensive option and the total cost, instead. When the user requests a quantity that is not feasible, let them know that their order quantity isn't feasible and present the least expensive option and total cost for the suggested alternative quantity. You can think of this as a second Diophantine equation. While you still need to identify the values that satisfy the quantity requirements of $6a + 9b + 22c = n$, there is an

additional constraint $3a + 4b + 9c = \text{cost}$ (where 3, 4, and 9 represent the costs associated with the 6, 9, and 22-piece options, respectively).